

Ethernet LAN Switching part-I

Day-5

29-03-2026

OSI model

7	Application Layer
6	presentation "
5	session
4	Transport
3	Network
2	Data Link
1	Physical

Physical Layer

→ Defines phy. characteristics of the medium used to transfer data between devices.

→ Ex:- voltage levels, maximum transmission distances, connectors, cable etc

→ Digital bits are converted into electrical or radio (wireless)

DATA Link

→ Node-to-node connectivity & data transfer (Ex: PC to switch, switch to router, router to router)

→ Define how data is formatted for transmission over a physical medium.

→ Detects & (possibly) corrects Physical Layer error

→ uses Layer 2 addressing separate from Layer 3 addressing

Switches are operate at L2

OSI model PDU

Data

Data L4 Header

Data L4 header L3 Header

L2 Header Data L4 Header L3 H L2 H

data

segment

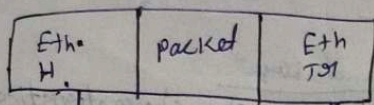
packet

frame

protocol Data units (PDUs)

Ethernet frame

frame check sequence



→ used by the Receiving device to detect any errors might have occurred in transmission.

Five fields in the header

FCS - 4 byte

1) Preamble SFD → Start Frame Delimiter

7 byte

1 byte

These are used for synchronization & to allow the receiving device to be prepared to receive the rest of the data in the frame.

2) Destination

6 byte

→ L2 address to which the frame is being sent

3) Source

6 byte

→ L2 address of the ^{device} which sent the frame

4) Type

2 byte

→ Indicate L3 protocol used in Encap. packet
IPv4, IPv6

Preamble

• Length: 7 byte (56 bits)

• Alternating 1's & 0's

• 10101010 * 7

• Allows device to synchronize their receiver clocks

SFD

• Length: 1 Byte (8 bits)

• 10101011

• Marks the end of the preamble & the beginning of the rest of the frame

Destination

Source

- Indicate the devices sending & receiving the frame
- Consist of the destination & source 'MAC' address
- MAC = Media Access control
- = 6 byte (48 bit) address of the physical device

Type

• 2 byte (16-bit) field

• A value of 1500 or less in this field indicates the LENGTH of the Encapsulated packet (in bytes)

or

Length

• A value of 1536 or greater in this field indicates the TYPE of the Encapsulated packet (usually IPv4 or IPv6), & length is determined via other methods

IPv4 = 0x0800 (hexadecimal)
(2048 in decimal)

IPv6 = 0x86DD (hexadecimal)
(34525 in decimal)

FCS

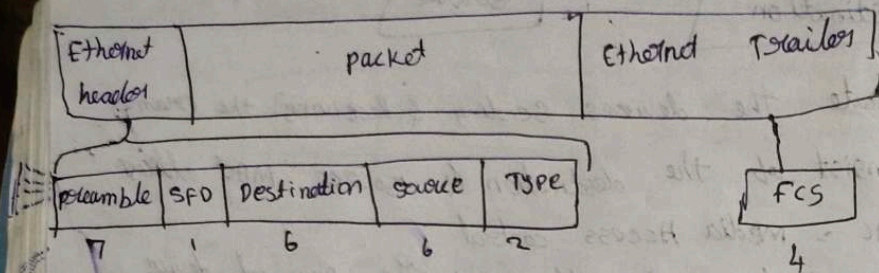
Frame check sequence

• 4 bytes (32 bits) in length

• Detects corrupted data by running a CRC algorithm over the received data.

• CRC = cyclic Redundancy check.

→ Redundancy Refers - to that fact that these 4 bytes at the end of the message enlarge the message without adding any new information.



$$22 + 4 = 26 \text{ bytes (header + trailer)}$$

MAC Address

=> 6 byte (48 bit) physical address assigned to the device when it is made

=> A-K-A Burned-In Address (BIA)

=> Is globally unique

=> The first 3 bytes are the OUI (organizationally unique identifier), which is assigned to the company making the device

=> The last 3 bytes are unique to the device itself.

=> written as 12 hexadecimal characters.

Decimal

-> use 10 possible digits: 0 1 2 3 4 5 6 7 8 9

100 means

1 value of 100
0 value of 10
0 value of 1

Hexadecimal

-> use 16 possible digits: 0 1 2 3 4 5 6 7 8 9 A B C D E F

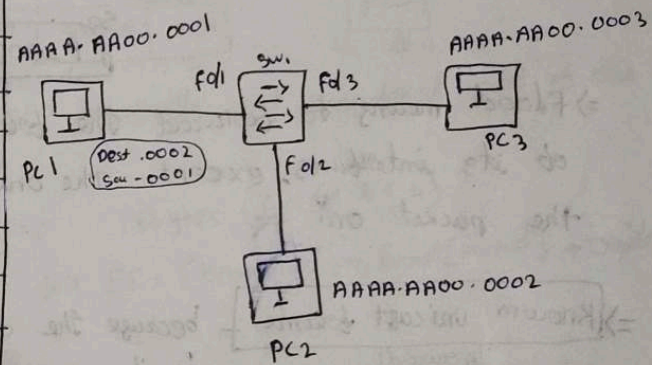
Dec	Hex
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F
16	10
17	11
18	12
19	13
20	14
21	15
22	16
23	17
24	18
25	19
26	1A
27	1B
28	1C
29	1D
30	1E

16 = this 10

Represents 1 value of 16
0 value of 1

17 = 11

1 value of 16
1 value of 1



Unicast frame

-> PC1 wants to send some data to PC2

-> A frame destined for a single target (PC2 in this case)

• PC1 sets a SW

Mac address Table	
MAC	Interface
0001	Fa1/24

Dynamically learned MAC

* This is how switches dynamically learn where each device on the network is, by looking at the source MAC address of the frame.

→ problem

⇒ The destination of the frame is AAAA.AA00.0002 but SW doesn't know where that is - This is called

unknown unicast frame

⇒ A frame for which the switch doesn't have an entry in its MAC address table.

So FLOOD the frame

⇒ Flood means to forward the frame out of ALL of its interfaces, except the one it received the packet on.

⇒ Known unicast frame - because the destination is already in its MAC address table

• whereas UNKNOWN unicast frames are flooded known unicast frames are simply forwarded to destination.

• Dynamic MAC addresses are removed from the MAC address table after 5 min of inactivity

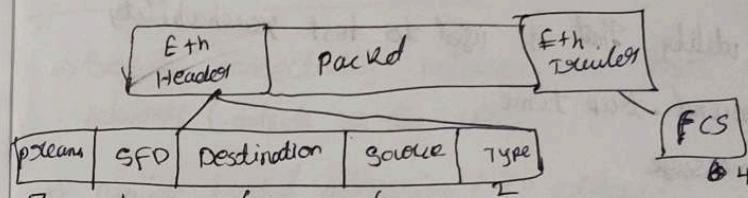
DAY - 6

07-04-2026

Ethernet LAN switching

Part - II

Ethernet Frame



⇒ The Preamble + SFD is usually not considered part of the Ethernet header

⇒ Therefore the size is 18 bytes (6 + 6 + 2 + 4)
⇒ minimum size for EF = (Header + payload [packet] + Trailer) is 64 bytes

⇒ 64 bytes - 18 bytes (header + trailer size) = 46 bytes

⇒ minimum payload (packet) = 46 bytes / if less than 46 padding bytes are added

⇒ 34 byte packet + 12 byte padding = 46

ARP (Address Resolution Protocol)

⇒ used to discover Layer 2 address (MAC address) of a known Layer 3 address (IP)

⇒ consists of 2 type messages

- ARP Request → send all hosts on the network
- ARP Reply → send only to one host (the host that sent the request)

ARP Table

• use arp -a | Type static - default entry
- Learned via ARP

Ping

- A network utility that is used to test reachability
- measures round-trip time
- use two message
 - ICMP Echo Req
 - ICMP Echo Reply
- Command ping (ip address)